

Task 1

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Tasks

- Solve the gridworld environment SeaShanty in file gridworld.py within as few episodes as possible (environment SeaShanty(size=12, period=6))
- Plot your results as the real cumulative reward of the current episode (y-axis) over episodes (x-axis), averaged over 100 different environments

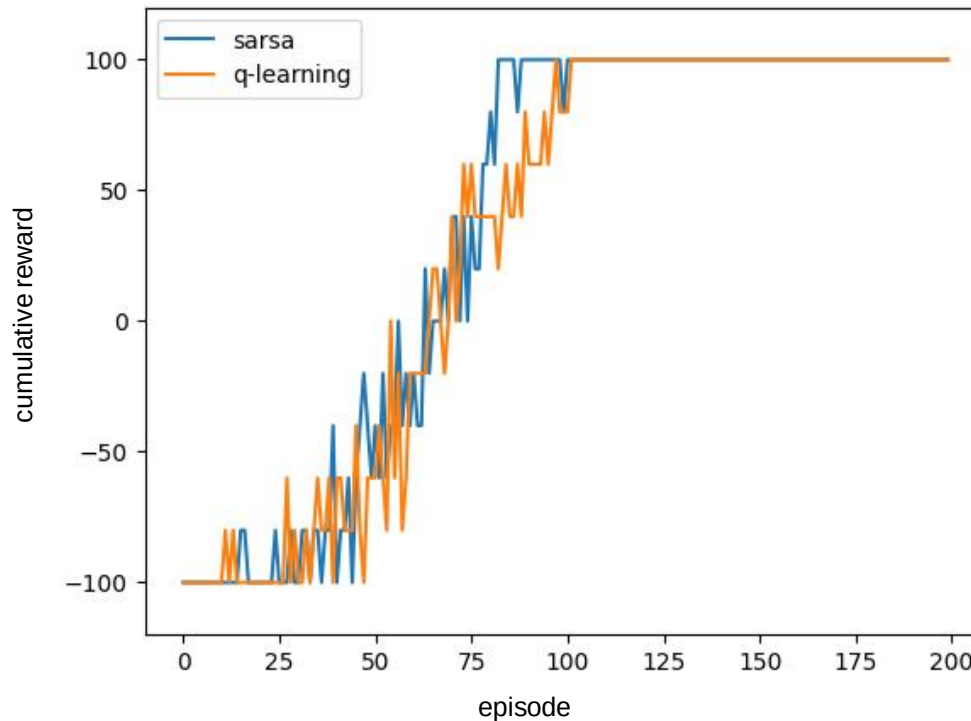
Deadline: 19.07.2026 23:59

Hints:

- You are allowed to use any public function of the environment, except for the step_dp function
- You are allowed to look at the existing code to understand how SeaShanty works
- Have a look at [Optuna](#) for optimizing your hyperparameters automatically (e.g. ϵ , α , ...)
- Do not use the same environments for hyperparameter optimization / final plots
- The real cumulative reward is the sum of all the "true" (not reward-shaped) rewards obtained within an episode

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- Your plots should roughly look as shown below. Please ensure that the resulting plots focus on the episodes where the environment is just about getting solved until it is solved consistently for the first few times.



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Notes on task 1:

- All solutions must be implemented by yourself (not in a group) in Python 3
- Do not modify any of the provided Python files
- Do not analyze the environment algorithmically (e.g. if a valid path can be found)
- Your Moodle submission includes the following:
 - your code (Jupyter notebook or Python file)
 - a readme.txt explaining how to run the code
 - a pdf/doc(x) file containing at most five pages (including images/plots) presenting the approaches you tried out, corresponding results and your final best solution.
You are encouraged to present approaches/ideas that did not work well, too.